1. Raspberry Pi (Generation 1)

The Raspberry Pi is a small ,affordable computer developed by the Raspberry Pi Foundation in the UK. It was created to help students learn programming and computer basics.

The first Raspberry Pi (Model B) was released in 2012. It had a 700 MHz ARM processor, 256 MB RAM, USB ports, HDMI output, and an SD card slot for storage. Even though it was small, it could run Linux and perform tasks like web browsing, programming, robotics, and home automation.

Today, Raspberry Pi is often used in education, IoT, AI projects, and embedded systems.

2. YOTTA

Yotta

Yotta Data Services is an Indian company that provides:Cloud computing services,AI infrastructure (including GPU AI training) ,Data center services,Disaster recovery

Its customers include businesses, startups, government organizations, and research institutions that need secure, large-scale computing infrastructure**.**

3. NPTEL

National Programme on Technology Enhanced Learning (NPTEL) is a free online learning platform created by the Indian Institutes of Technology and the Indian Institute of Science.

It offers: Free video lectures ,Online courses in engineering, science, management, humanities, and programming , proctored certification exams (paid) that can earn us certificates recognized by employers and universities

Popular subjects include:Python and Java programming ,Machine Learning ,Data Science ,Computer Networks ,Operating Systems ,Digital Marketing , Project Management

4. 10 key takeaways from the recent AI Impact Summit 2026

1. A coalition of 88 nations has come together, signing the New Delhi Declaration to ensure that AI is secure, trustworthy, and accessible to all, not just affluent countries.
2. A significant commitment of over $250 billion has been made by corporations and governments to develop AI infrastructure, with major investments from industry leaders like Reliance, Google, and Microsoft.
3. The emphasis has shifted from merely discussing AI safety to implementing AI technologies effectively in areas such as healthcare, agriculture, and education.
4. India introduced its first domestically developed advanced AI models from Sarvam AI, which support local Indian languages and are freely available for public use.
5. Leading technology firms have disclosed that AI "agents" are now responsible for generating approximately 80% of their code.

6. 6. Ending Monopolies: In order to encourage fair competition and open-source tools, nations decided to prevent a small number of powerful corporations or states from dominating all AI technology.
7. Assisting Developing Countries: One of the main objectives was to assist developing nations in catching up by exchanging information, resources, and instruction via new international "commons" or libraries.
8. Cybersecurity Challenges: Leaders cautioned that although AI aids in the discovery of security flaws, it also speeds up and lowers the cost of cyberattacks, necessitating ongoing human attention.
9. Green AI: As AI systems get bigger, there has been a significant drive to make them more ecologically friendly and energy-efficient.
10. Humans Remain in Charge: Despite AI's speed, experts concur that human judgment is indispensable for morality, critical decision-making, and innovative problem-solving.

5. How to decide which programming language to use for an application

The programming language depends on the type of application you want to build.

Web development: JavaScript, Python, PHP

Android apps: Kotlin or Java

iOS apps: Swift

AI and Machine Learning: Python

System software: C and C++

Game development: C++ and C#

Data analysis: Python and R

Before choosing a language, we should consider:

Performance

Difficulty

Available libraries

Platform compatibility

Development time

6. RAM size, Word size, and 64-bit system

RAM(Random Access memory) is the temporary memory used by the computer to store data while programs are running. More RAM allows more applications to run smoothly.

Word size is the amount of data the CPU processes at one time. Most word size today are 32-bit and 64-bit.

64-bit : The processor handles 64 bits of data at once.

It can use much more than 4 GB of RAM and perform better for modern software.

It supports larger calculations and multitasking.

7. Why is a GPU faster than a CPU?

A CPU has a few powerful cores designed to perform different tasks one after another.

A GPU has thousands of smaller cores designed to perform many similar tasks at the same time.

Because of this parallel processing, GPUs are much faster for graphics rendering, AI, scientific calculations, and video processing.

8. CPU vs GPU vs TPU

CPU (Central Processing Unit):

- General-purpose processor
- handle operating system and daily tasks
- Few powerful cores

GPU (Graphics Processing Unit)

- Designed for graphics and parallel computing
- Thousands of smaller cores
- Used in gaming, AI, and video editing

TPU (Tensor Processing Unit)

- Special processor developed by Google
- Built specifically for AI and machine learning
- Faster and more power-efficient for neural network calculations

9. Last year's Turing Award

In 2025, Turing Award was given to Andrew G. Barto and Richard S. Sutton for developing the foundations of reinforcement learning, which is used for training models.

10. Stampeding Herd Problem

The Stampeding Herd Problem occurs when many users or processes try to access the same resource at the same time.

Example: if a popular website's cache expires, thousands of users may request the same data from the database simultaneously. This can overload the server and reduce performance.

This problem can be reduced using caching, request queuing, locking, and load balancing.

11. Founder of Intel and AMD

Intel: Robert Noyce and Gordon Moore

AMD: Jerry Sanders along with seven co-founders

12. Which CPU organization is used in your laptop?

My laptop uses General Register Organization primarily and the CPU performs calculations using 16 architectural general-purpose registers. It also uses Stack Organisation However, this stack is used primarily for control flow (function calls, returns, local variables) and temporary data storage, not as the main mechanism for arithmetic.